

MS MARCO Generative QA — Progress Report

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Project goal

Build an end-to-end open-domain QA system on MS MARCO that goes **query** → **retrieve passages** → **generate an answer**, and measure each stage in isolation so it's clear where errors come from. The aim is not SOTA; the aim is a clean, reproducible pipeline whose retrieval, reranking, and generation stages can each be ablated.

Methods (by week)

Week	Stage	Data	Method
W1	EDA	MS MARCO QA v2.1 validation, 5k-row sample	Length / type / answer-type distributions; sizing decisions for W2–W5
W2	Sparse retrieval (BM25)	MS MARCO Passage (8.8M docs), dev/small (6 980 q)	bm25s backend, $k_1=1.5$, $b=0.75$, top-1000 per query
W3	RAG generation	BM25 / reranked top-3 $\cap$ QA references, full dev/small (6 980 q)	T5-small (no fine-tuning), prompt = question: <q> context: <p1 p2 p3>
W4	Dense retrieval	Qrels-anchored 50k sample of Passage, dev/small (6 980 q)	all-MiniLM-L6-v2 + FAISS IndexFlatIP; same sample re-indexed with BM25 for apples-to-apples comparison
W5	Cross-encoder reranking	W4 dense top-100, full dev/small (6 980 q)	cross-encoder/ms-marco-MiniLM-L-6-v2 rerank, depth K=100
W6	Evaluation layer + decode-budget closure	W3 paired predictions (6 980 q), 3 000-pair BERTScore subsample, 40-query	DistilBERT BERTScore proxy; deterministic 5-bucket regression taxonomy; max_new_tokens 64→128 falsification run; question-form / structural-feature breakdowns (W6-A/B/C)

Week	Stage	Data	Method
		taxonomy triage, mnt=128 sweep	
W7	Grounding audit	W3 paired predictions (6 980 q), 3 000-pair NLI subsample	Lexical content-token + 3-gram grounding (deterministic CPU, ~2 s); DeBERTa-v3 NLI entailment proxy; grounding [?] Token-F1 / BERTScore correlation; low- grounding case study (W7-A/C/ D)
W5- A	First-stage × rerank head- to-head	BM25 full-corpus top-100 + W4 dense top-100 (both 6 980 q)	Same ms-marco-MiniLM-L-6- v2 reranker on both first stages; reports Δ and constrained recovery rate

All runs share a single config (configs/baseline.yaml); the four entrypoints in experiments/run_*.py produce the W1–W5 metrics each report cites. W6 and W7 are *post-hoc analysis layers* over the already-on-disk W3 predictions — no new generation or model load, all under scripts/*.py.

Key results

W1 (EDA) — Queries are short (5–10 words median); passages ~10× longer (median ~50 words). DESCRIPTION and NUMERIC dominate the query-type buckets; a non-trivial fraction of queries has no answer and must be filtered for any SFT supervision. Most queries have 0 or 1 marked-relevant passage (sparse, binary qrels).

W2 (BM25 baseline, full 8.8M corpus, 6 980 dev/small queries)

Metric	Value
MRR@10	0.1703
Recall@100	0.6212
Recall@1000	0.8154

Published Anserini/Lucene reference is ~0.184 MRR@10; the ~0.014 gap is attributable to tokenizer differences between Lucene and bm25s. No k1/b tuning. Search cost: 586 ms/query.

W3 (RAG, T5-small + retrieved top-3, full dev/small 6 980 queries)

Two paired runs over the **same 6 980 dev/small qids** (mutually restricted via --restrict-to-run), differing only in upstream retrieval source:

Retrieval → T5-small	ROUGE-L	BLEU	EM	Token-F1
BM25 top-3	0.1859	0.0717	0.0135	0.1966
Reranked top-3	0.3621	0.2922	0.0606	0.3677
Δ (rerank – BM25)	+0.1763	+0.2206	+0.0471	+0.1711

Reranking the first stage **roughly doubles every generation metric** on the full benchmark. 95 % paired-bootstrap CIs (n = 6 980, 10 k resamples, seed 42) on the per-query Δ all sit strictly above zero: ROUGE-L [+0.1663, +0.1820], BLEU [+0.1265, +0.1395], EM [+0.0417, +0.0527], Token-F1 [+0.1632, +0.1789]; two-sided $p < 0.001$ in every case.

Mechanism (retrieval flag): the rate of having at least one relevance-judged passage in the top-3 jumps from **20.8 % (BM25) to 96.9 % (reranked)** — the cross-encoder is what gets the relevant passage into the generator's window.

By query type (token-F1 Δ), DESCRIPTION (n=3725) gains most (+0.2050); NUMERIC (n=1665) gains least (+0.1238) — short numeric answers depend more on lexical surface match than on which-of-the- near-duplicates the reranker picks. Per-query split: **4 015 improvements / 1 766 regressions / 1 199 ties** over 6 980 queries.

T5-small is pretrained-only (no SFT) so absolute level numbers are modest. The load-bearing claim is the **paired Δ between rows**, which is statistically robust on n = 6 980.

*A low-cost semantic evaluation sanity check on 3 000 paired examples using DistilBERT-based BERTScore (rescaled, paired-bootstrap CI; seed 42) gives BM25 0.2192 $\rightarrow$ reranked 0.3920, Δ +0.1728 ([+0.1608, +0.1850], $p < 0.001$). Per-query: rerank strictly better on 64.8 % of qids; tie 5.7 %; strictly worse on 29.5 %. The semantic-proxy Δ (+0.1728) is within a hair of the surface-form Token-F1 Δ (+0.1711) and ROUGE-L Δ (+0.1742), so the reranker improvement **is not a surface-form artefact** — it shows up in a semantic-similarity scorer at the same magnitude. (DistilBERT is intentionally not the canonical BERTScore encoder; this is a proxy that complements rather than replaces a full roberta-large evaluation.)*

A 40-query seeded triage of the 233-strong regression bucket shows that **~90 % of regressions are generation-side truncation, not retrieval or semantic failures**: 55 % truncation_midword, 35 % truncation_short (≤ 3 tokens, generator extracted only a title- like fragment), 5 % topic_drift, 5 % extractive_passage_bias, 0 % semantic_mismatch. The midword pattern initially looked like the max_new_tokens = 64 cap firing; W6's closure run below **falsifies** that reading.

W4 (Dense vs BM25 on the same 50k qrels-anchored sample, 6 980 queries)

Metric	BM25 (sample)	Dense (sample)	Δ
MRR@10	0.6948	0.8830	+0.1882
nDCG@10	0.7270	0.9041	+0.1771
Recall@100	0.9338	0.9946	+0.0608
Recall@1000	0.9686	0.9991	+0.0305

Dense wins on every metric. Caveat: every dev/small relevant doc is unconditionally included in the 50k pool, so absolute numbers are upper-bounded; the *delta* between dense and BM25 on the same pool is what's meaningful. Recall@100 = 0.9946 means dense has effectively hit the ceiling on this sampled setting $\rightarrow$ from W5 onward, recall is no longer the bottleneck.

W5 (Cross-encoder rerank over W4 dense top-100, full 6 980-query dev/small)

Metric	Dense (top-100)	Dense + CE rerank	Δ
MRR@10	0.8830	0.9304	+0.0474
nDCG@10	0.9041	0.9434	+0.0393
Recall@100	0.9946	0.9946	+0.0000

Recall@100 is unchanged by construction (the reranker only re-orders the top-100). The interesting columns are MRR@10 and nDCG@10: even with recall saturated, the cross-encoder still buys roughly +0.04 by tightening *local ordering* — pushing the relevant passage from rank 2–3 to rank 1. Cost: 4 h 37 min wall-clock for 538 k pairs (~32 pairs/s), peak RSS 3.3 GiB; the CE is the slowest stage by ~2 orders of magnitude. The earlier 1 000-query subsample of dev/small produced consistent deltas (MRR Δ +0.0435, nDCG Δ +0.0398), confirming the gain is not an artefact of the subsample.

W5-A (Rerank head-to-head across first stages, same 6 980 dev/small queries)

The same ms-marco-MiniLM-L-6-v2 reranker applied to two first stages of very different starting strength:

First stage	MRR@10 (FS)	MRR@10 (+ CE)	Δ_{MRR}	nDCG@10 (FS)	nDCG@10 (+ CE)	Δ_{nDCG}	Constrained MRR recovery
BM25 (full 8.8M)	0.1703	0.3554	+0.1851	0.2138	0.4017	+0.1879	41.1 %
Dense MiniLM-L6 (50k)	0.8830	0.9304	+0.0474	0.9041	0.9434	+0.0393	42.4 %

Absolute deltas favour the BM25 arm (more room to recover), but the **constrained recovery rate** — $\Delta / (\text{Recall@100} - \text{first-stage})$ — comes in tied near ~42 % MRR / ~44 % nDCG once you account for the recall ceiling of each arm (BM25 full-corpus Recall@100 = 0.6212 caps what the reranker can promote; dense-on-sample sits at 0.9946). Read together with W2 + W4 + W5: the cross-encoder closes a fixed *fraction* of the ordering gap, regardless of first-stage strength.

W6 (Evaluation layer — semantic proxy, regression taxonomy, decode-budget closure)

Layered on the W3 paired predictions, no new generation:

- *Semantic-proxy BERTScore* on a 3 000-pair subsample (DistilBERT, rescaled, paired bootstrap 10 k resamples seed 42): BM25 0.2192 $\rightarrow$ reranked 0.3920, Δ **+0.1728 [+0.1608, +0.1850], $p < 0.001$** . Per-query: rerank strictly better 64.8 %, tie 5.7 %, BM25 better 29.5 %. Lands within ± 0.002 of the surface-form Token-F1 Δ (+0.1711) and ROUGE-L Δ (+0.1742) — so the rerank gain is **not a surface-form artefact**.
- *Regression-bucket triage* (40-query seeded sample of the 233-strong regression bucket): 90 % generator-side truncation as summarised under W3 above.
- *Decoding-budget closure* ($\text{mnt} = 64 \rightarrow 128$, full dev/small): same generator, same prompts, same retrieval inputs, same seed; only the decode cap changes.

Result: Δ Token-F1 +0.1706 (CI [+0.163, +0.178]), Δ ROUGE-L +0.1736, Δ BLEU +0.1325, Δ EM +0.0471 — all four deltas move by < 0.005 , the regression bucket shrinks 233 $\rightarrow$ 231, and the 40-query truncation share moves 90.0 % $\rightarrow$ 87.5 %. **The W6 budget-cap hypothesis is falsified:** T5-small is hitting EOS naturally on this prompt format, not running out of decode budget. The midword-ending style is intrinsic to the model.

- W6-A/B/C offline breakdowns: question-form tagger over all 6 980 queries; perform Δ Token-F1 (only which, $n = 120$, has a CI on Δ Token-F1 that includes zero — $\Delta = +0.0250$, CI [-0.028, +0.077], $p = 0.33$, despite its retrieval-side Δ MRR@10 = +0.71); Mann-Whitney U on five structural features of regression vs non-regression queries (only signal: regression-arm top-3 passage lengths are marginally shorter, Δ median = -2.7 tokens, $p = 0.0073$, $r = -0.103$ — consistent with the truncation finding).

W7 (Grounding audit — what is T5-small actually doing?)

Deterministic CPU pass over the same W3 paired predictions; ~2 s of scoring + bootstrap, no model load except for the optional NLI proxy.

- *Lexical content-token grounding* (fraction of unique non-stopword prediction tokens appearing in the prompt's top-3 passages): BM25 **0.9972** $\rightarrow$ reranked **0.9977**, Δ **+0.0005** [-0.0003, +0.0014], $p = 0.24$. Both arms at the ceiling.
- *3-gram grounding*: BM25 **0.9873** $\rightarrow$ reranked **0.9905**, Δ **+0.0032** [+0.0015, +0.0050], $p < 0.001$. Slight uplift, both arms still ~99 %.
- **Read:** T5-small on this prompt format is performing **extractive QA**. The W3/W5 Δ Token-F1 = +0.171 is almost entirely downstream of retrieval — the reranker puts the right words into the prompt and the generator copies them. This is a *calibration* of the W3 result, not a contradiction.
- W7-A — *NLI-entailment grounding* (3 000-paired subsample, cross-encoder/nli-deberta-v3-small): BM25 **0.2270** $\rightarrow$ reranked **0.0821**, Δ **-0.1448** [-0.1597, -0.1297], $p < 0.001$. The only metric in this project whose paired Δ **reverses sign** versus the W3/W5/W6 surface-form story. Likely mechanism: reranked-arm predictions are more fragmentary / mid-word-cut, which inflates word and 3-gram overlap but scores low on a sentence-level NLI cross-encoder that cannot entail a fragmentary hypothesis. To be tested by a generator swap (W7-B follow-up: T5-base / instruction- tuned small model).
- W7-C — *grounding [?] downstream correlation*: every binned cell (high-vs-low grounding Mann-Whitney) shows downstream-better-on- high direction, but magnitudes are 0.04–0.10 in absolute Token-F1 / BERTScore — consistent with the ceiling effect. Per-query Spearman / Pearson are near-zero, again because almost all rows sit ≥ 0.99 on lex / 3-gram.
- W7-D — *low-grounding case study* (30-query seeded triage of the 197 rerank-arm queries with lex < 0.9 OR ngram < 0.9): 77 % paraphrase_reorder, 23 % partial_external (mostly tokeniser / morphology artefacts: 350oF vs 350°F; competed vs compete), **0 % parametric-or-external hallucinations**. Reinforces W7's headline: T5-small is extractive even on its worst-grounded outputs.

Limitations

- **Sampled corpus from W4 onward.** Dense + rerank numbers are on a 50k qrels-anchored sample where every relevant doc is present by construction.

Comparable *within* W4/W5; the W4/W5 absolute MRR is *not* directly comparable to the W2 full-corpus number.

- **No fine-tuning anywhere.** Generic encoders/generator used as-is (all-MiniLM-L6-v2, ms-marco-MiniLM-L-6-v2, t5-small). A MS-MARCO-tuned encoder (msmarco-MiniLM-L6-cos-v5) or SFT generator would close most of the gap to leaderboard baselines.
- **Dev/small only, single seed.** No held-out test split; distractor sample is deterministic but not averaged across seeds.
- **No fine-tuning of T5-small.** W3 absolute level numbers (ROUGE-L ~0.18 BM25 → ~0.36 reranked) are modest because the generator is pretrained-only; the Δ between paired rows is the load-bearing signal. A SFT pass on (question, gold passage, answer) triples is in scope for a future week.
- **Surface-form generation metrics only at W3.** The W6 BERTScore proxy (semantic-similarity) and the W7 grounding metrics (extractiveness) partially close this gap, but the canonical roberta-large BERTScore on full dev/small is still open.
- **NLI-grounding sign flip on the rerank arm (W7-A) is unresolved.** Direction reversal is statistically robust (CI strictly above zero in magnitude); the mechanism is hypothesised — not yet confirmed — to be the fragmentary-output artefact rather than a true semantic regression. A generator swap is needed to disentangle.
- **No fine-tuning anywhere in the pipeline.** Every model is used as a generic pretrained checkpoint; no SFT pass on (question, gold passage, answer) triples has been done. This bounds how far the surface-form / semantic deltas can go.

Next steps

The W6 closure + W7 grounding audit together re-aim the remaining follow-up work away from the decode budget and toward the generator itself and the prompt format:

- **W7-B — generator capacity, not decode budget.** The closure run
 - the 99 % extractiveness ceiling rule out the `max_new_tokens = 128` intervention. The candidate axes are: (i) swap T5-small for a larger or instruction-tuned variant on the same prompts (T5-base is the planned first step under the reduced-scope ddl); (ii) move off the question: ... context: ... extractive-QA shape toward a prompt that demands multi-passage synthesis or explicit citation; (iii) light SFT on (question, gold passage, answer) triples — cheap to try, but the W7 ceiling suggests gains will be bounded by the prompt shape, not by lack of supervision.
- **Resolve the W7-A NLI sign flip.** The reranked arm's NLI Δ is the only metric reversing direction in the whole project; the current best guess is that fragmentary midword outputs score low on a sentence-level cross-encoder even when lexically grounded. A T5-base swap on the same prompts would disambiguate "generator fragment artefact" from "true semantic regression".
- **Citation-grade BERTScore.** Re-run the W6 proxy with roberta-large on full dev/small (~hours of CPU). The proxy and the surface-form deltas agree at the same magnitude, so this is a paper-writing requirement, not a research question.
- **W4 follow-ups.** W4-B: head-to-head on the same 50 k qrels- anchored sample for bge-small-en-v1.5, all-MiniLM-L12-v2, and the existing all-MiniLM-

- L6-v2 baseline (the L12 run is on disk; BGE is queued). *W4-A*: relevant-doc-density sensitivity (5 % / 10 % only, per ddl scope) with the W4-B winner.
- **W5-B — $K \in \{50, 100, 200\}$ sweep** on both first stages for the performance-latency Pareto. Subsample, per ddl scope.
 - **Single-query demo CLI** (polish tier). The pipeline is currently batch-eval-oriented; a thin python `-m src.demo.ask "<question>"` wrapper that bundles the doc_id $\rightarrow$ passage map would make the end-to-end system approachable without running a full retrieval sweep.

How this maps to the repo

- Conclusions per week: `reports/generated/weekNN_*.pdf`
- Process logs per week: `notebooks/weekNN_*.ipynb`
- Reproducible pipeline: `experiments/run_*.py` + `src/` + `configs/baseline.yaml`
- Reproduce locally: `make` targets (see `Makefile`); outputs land in `outputs/` (not committed).